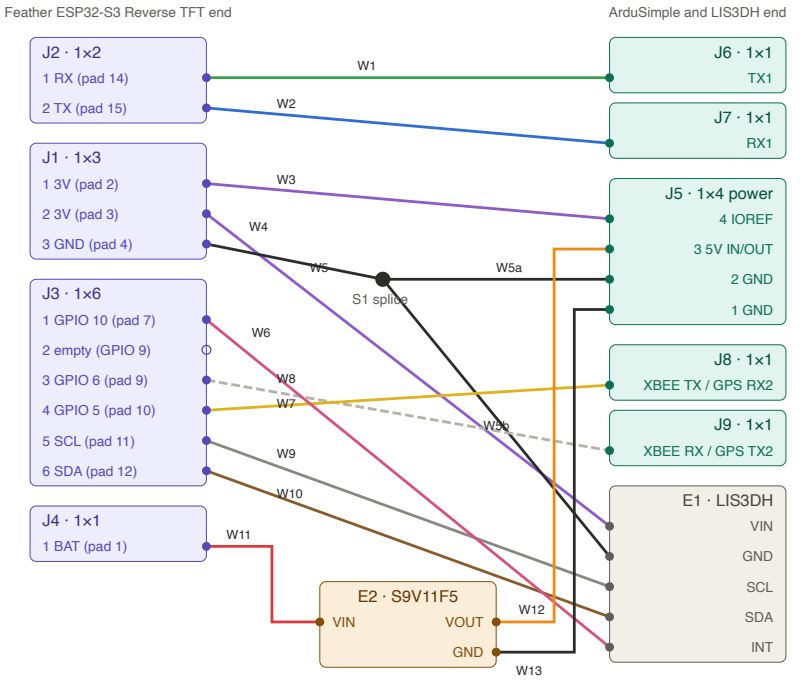
ESP32 rover wiring harness

Request for quote. One harness and two RF pigtails per boat. Rev A, 2026-11-05.

1. Harness drawing



J = 2.54 mm crimp housing (female) onto male header pins. E = tinned leads, soldered. Pad numbers count from USB-C end.

Connectors

Ref	Plugs onto	Housing	Positions
J1	Feather left edge, pads 2–4	1×3, 2.54 mm, female crimp	3V, 3V, GND. Pad 1 (RST) must stay uncovered.
J2	Feather left edge, pads 14–15	1×2, 2.54 mm, female crimp	RX (GPIO 38), TX (GPIO 39)
J3	Feather right edge, pads 7–12	1×6, 2.54 mm, female crimp	10, *empty*, 6, 5, SCL, SDA
J4	Feather right edge, pad 1	1×1, 2.54 mm, female crimp	BAT
J5	ArduSimple power header	1×4, 2.54 mm, female crimp	GND, GND, 5V IN/OUT, IOREF. The WT pin next to it stays uncovered.
J6–J9	ArduSimple TX1, RX1, XBEE TX, XBEE RX pins	1×1 each, 2.54 mm, female crimp	One signal each
E1	Adafruit LIS3DH breakout	none: tinned leads	Strip 3 mm, tin. Soldered to VIN, GND, SCL, SDA, INT.
E2	Pololu S9V11F5 regulator	none: tinned leads	Strip 3 mm, tin. Soldered to VIN, VOUT, GND. Its EN pin stays unconnected.

Pad numbering: Feather left edge has 16 pads (RST, 3V, 3V, GND, A0…, RX, TX, last); right edge has 12 (BAT, EN, USB, 13, 12, 11, 10, 9, 6, 5, SCL, SDA), both counted from the USB-C end.

2. Wire list

Wire	From	To	Signal	Colour	Length
W1	J6 TX1	J2-1 RX	GPS data (UBX) to the ESP32	green	120 mm
W2	J2-2 TX	J7 RX1	commands and corrections to the GPS	blue	120 mm
W3	J1-1 3V	J5-4 IOREF	3.3 V logic reference	violet	120 mm
W4	J1-2 3V	E1 VIN	3.3 V to accelerometer	violet	80 mm
W5	J1-3 GND	S1 splice	ground	black	20 mm
W5a	S1 splice	J5-2 GND	ground	black	100 mm
W5b	S1 splice	E1 GND	ground	black	60 mm
W6	E1 INT	J3-1 GPIO 10	motion interrupt	pink	80 mm
W7	J8 XBEE TX	J3-4 GPIO 5	radio data in	yellow	120 mm
W8	J3-3 GPIO 6	J9 XBEE RX	radio data out	white	120 mm
W9	J3-5 SCL	E1 SCL	I²C clock	grey	80 mm
W10	J3-6 SDA	E1 SDA	I²C data	brown	80 mm
W11	J4 BAT	E2 VIN	battery, 3.0–4.2 V, ~0.3 A	red	60 mm
W12	E2 VOUT	J5-3 5V IN/OUT	5 V to the GPS board, ~0.15 A	orange	100 mm
W13	E2 GND	J5-1 GND	ground	black	100 mm

Length is end to end: from the face of one housing (or the end of the tinned lead) to the next. Tolerance ±5 mm.

Build requirements

Item	Requirement
Wire	26 AWG stranded tinned copper, UL1007 or silicone. The same gauge for every wire.
Terminals	2.54 mm female crimp ("Dupont"), gold flash, rated for 26 AWG. One wire per terminal.
Splice S1	Solder or crimp splice, covered with heat-shrink. Place it about 20 mm from J1.
Labels	Each connector and each tinned end marked with its reference (J1…J9, E1, E2), by a heat-shrink marker or a printed flag.
Bundling	Twist W1 with W2 and W8 with W9. The rest may be loose or sleeved.
Test	Continuity test of every wire against this list. No shorts between any two nets.

Quote details

Quantity: _____ harnesses, plus _____ prototype sets for a fit check first.

Please quote unit price at each quantity, tooling or setup charge, and lead time.

3. RF pigtails

Two per boat. Each runs from a jack on a board inside the case to a bulkhead in the case wall, where the antenna screws on. **RF1 is standard SMA; RF2 is reverse-polarity SMA (RP-SMA).** They must not be swapped.

	RF1 · GPS antenna	RF2 · radio antenna
Board end	SMA plug (male), straight. Mates with the ArduSimple simpleRTK2B's SMA jack.	RP-SMA plug, straight. Mates with the Digi XBee's RP-SMA jack.
Wall end	SMA jack (female) bulkhead, rear-mount, with nut and lock washer, O-ring sealed	RP-SMA jack bulkhead, rear-mount, with nut and lock washer, O-ring sealed
Antenna that screws on	GNSS helix with an SMA plug	XBee whip with an RP-SMA plug
Cable	RG316, 50 Ω	RG316, 50 Ω
Length	100 mm	100 mm
Frequency	1.2–1.6 GHz (GNSS L1/L2)	XBee band
Wall thickness	_____ mm. The bulkhead thread must pass through this with the O-ring and nut.	
Label	"GPS · SMA"	"RADIO · RP-SMA"

Length is from the plug's mating face to the bulkhead flange. Tolerance ±5 mm.